Universidad de La Laguna

Continuous Integration (CI), Vite, Github Actions and Github pages.

Continuous Integration (CI), Vite, Github Actions and Github pages.

Link to code examples in these slides

Authors

Marco Pérez Padilla

marco.perez.padilla.34@ull.edu.es

Pablo Aswani García

pablo.aswani.16@ull.edu.es

Iván Luis Estévez

ivan.luis.estevez.24@ull.edu.es

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What is Continuous integration?

Let’s see an example without continuous integration

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Let's see an example without continuous integration

Developer A

Developer B

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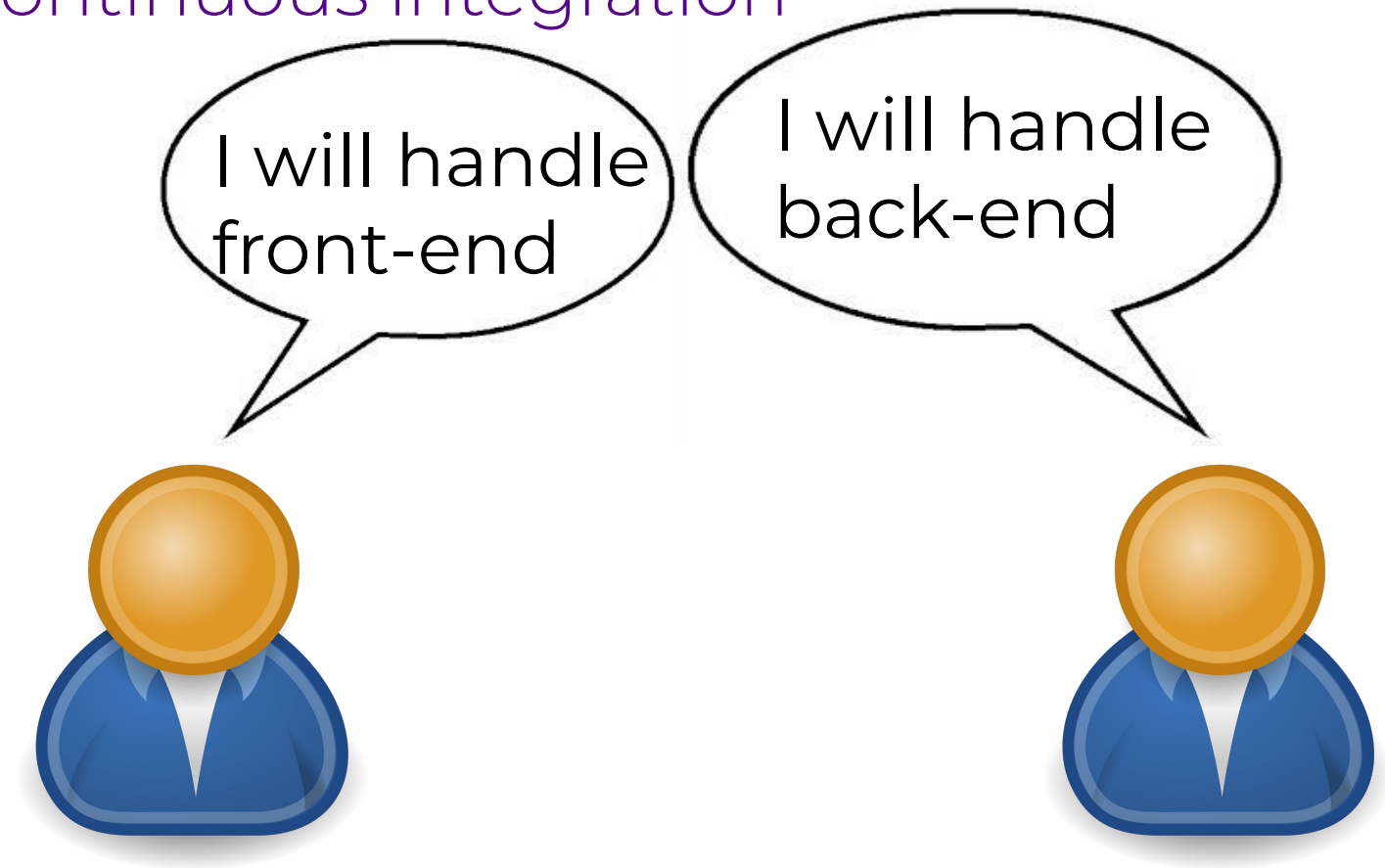
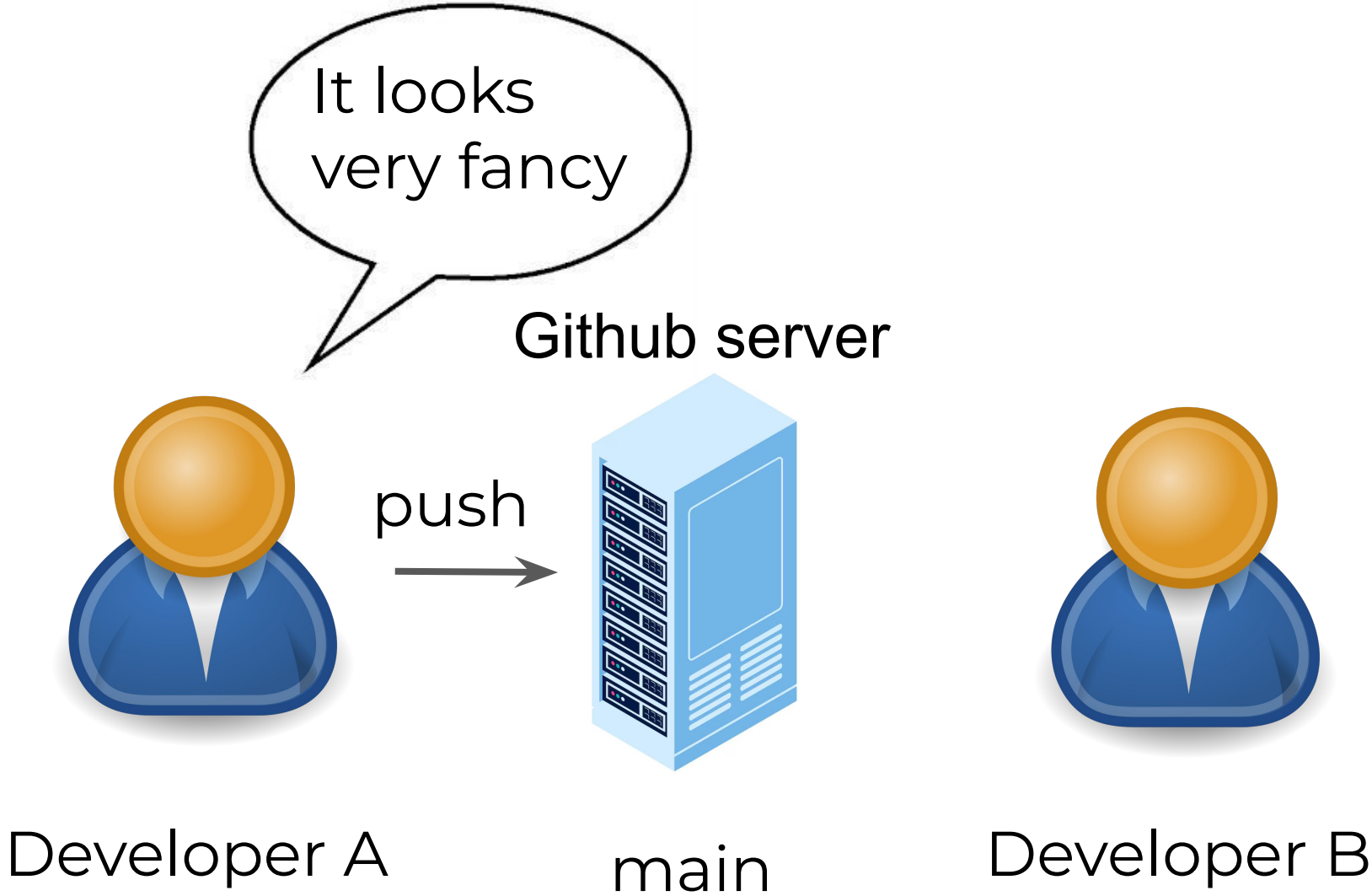
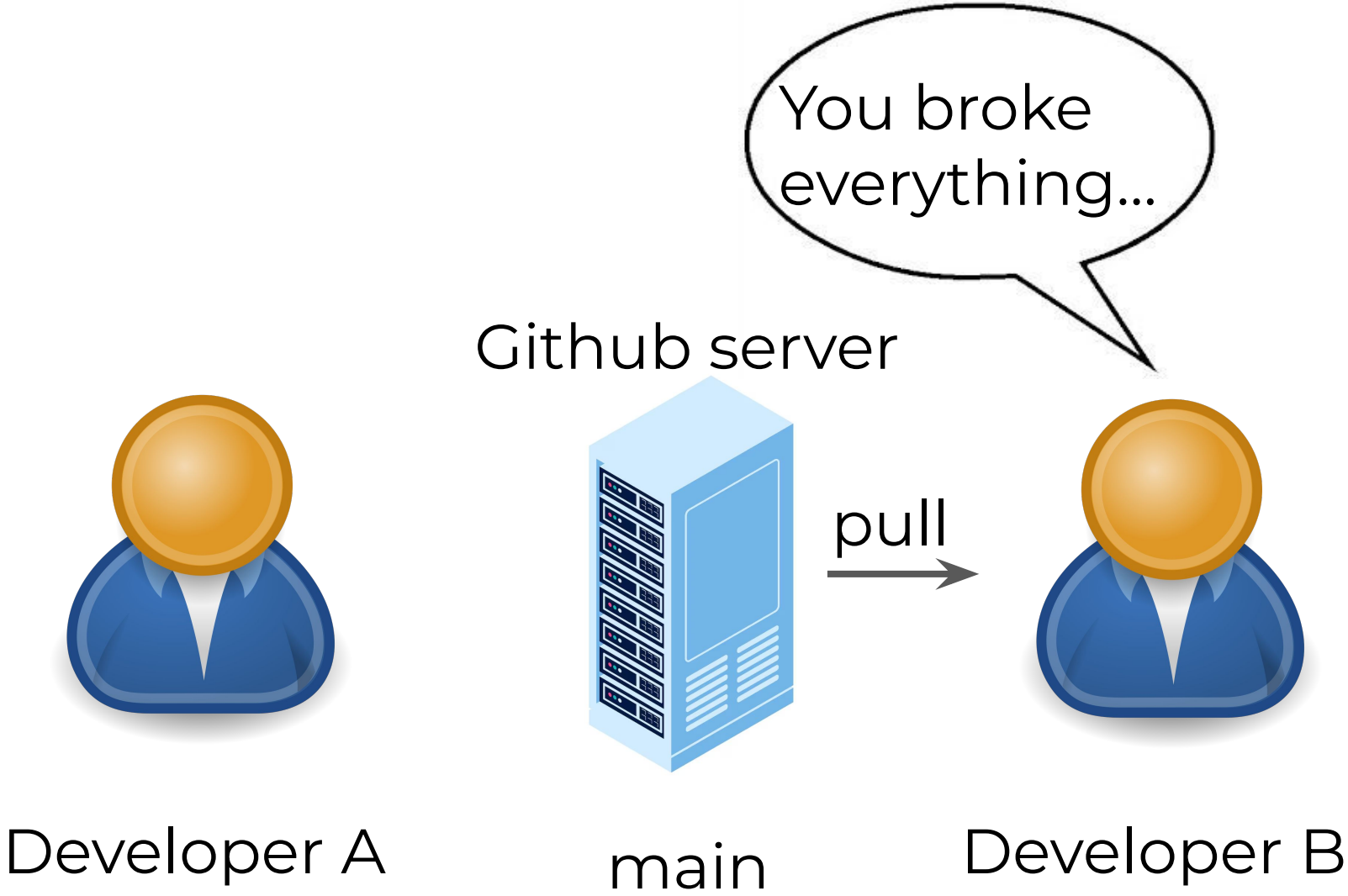
Let's see an example without continuous integration




It looks very fancy
Github server
push
Developer A
main
Developer B


You broke everything...
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pull
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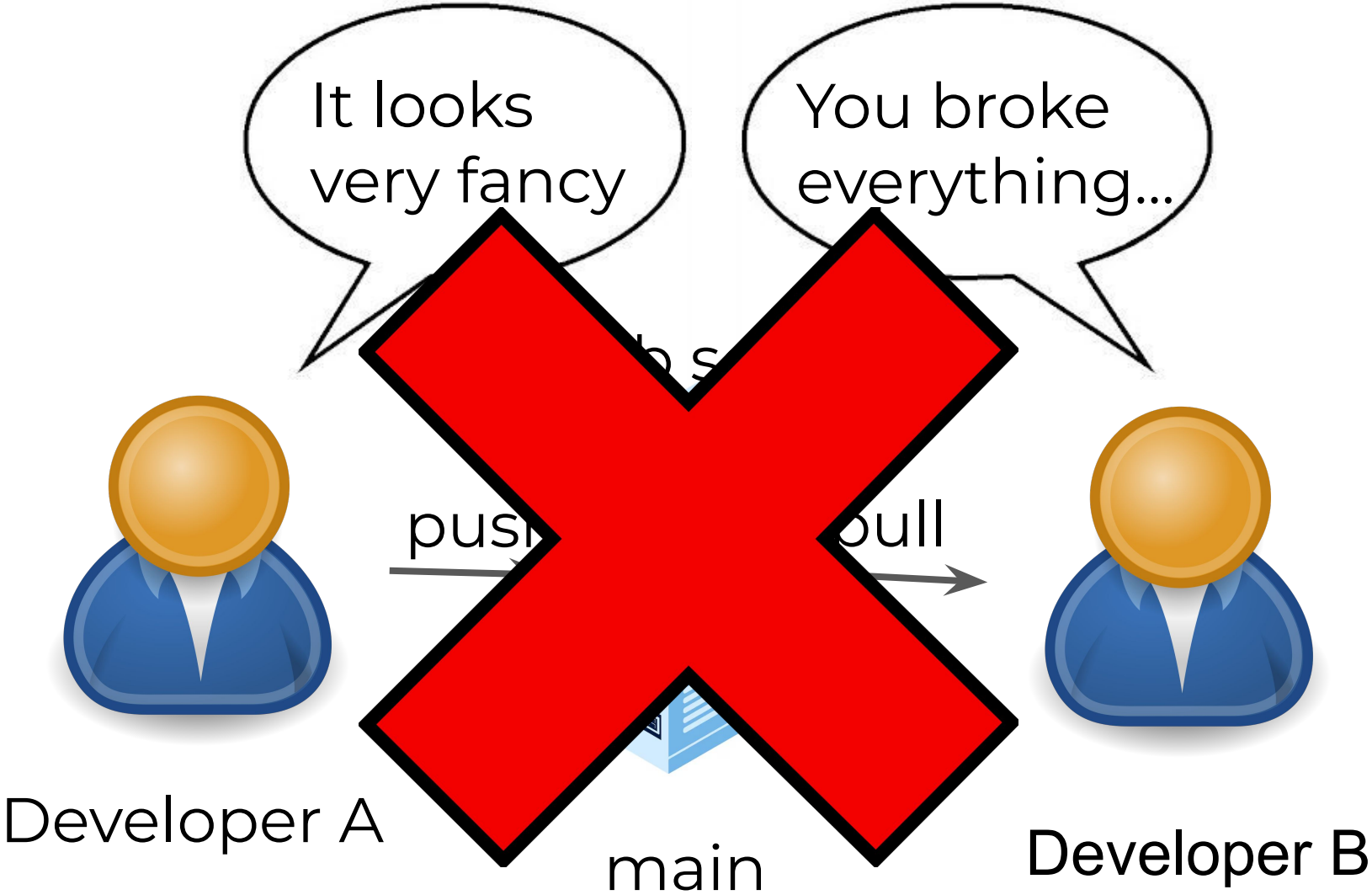
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You broke everything...
Developer A
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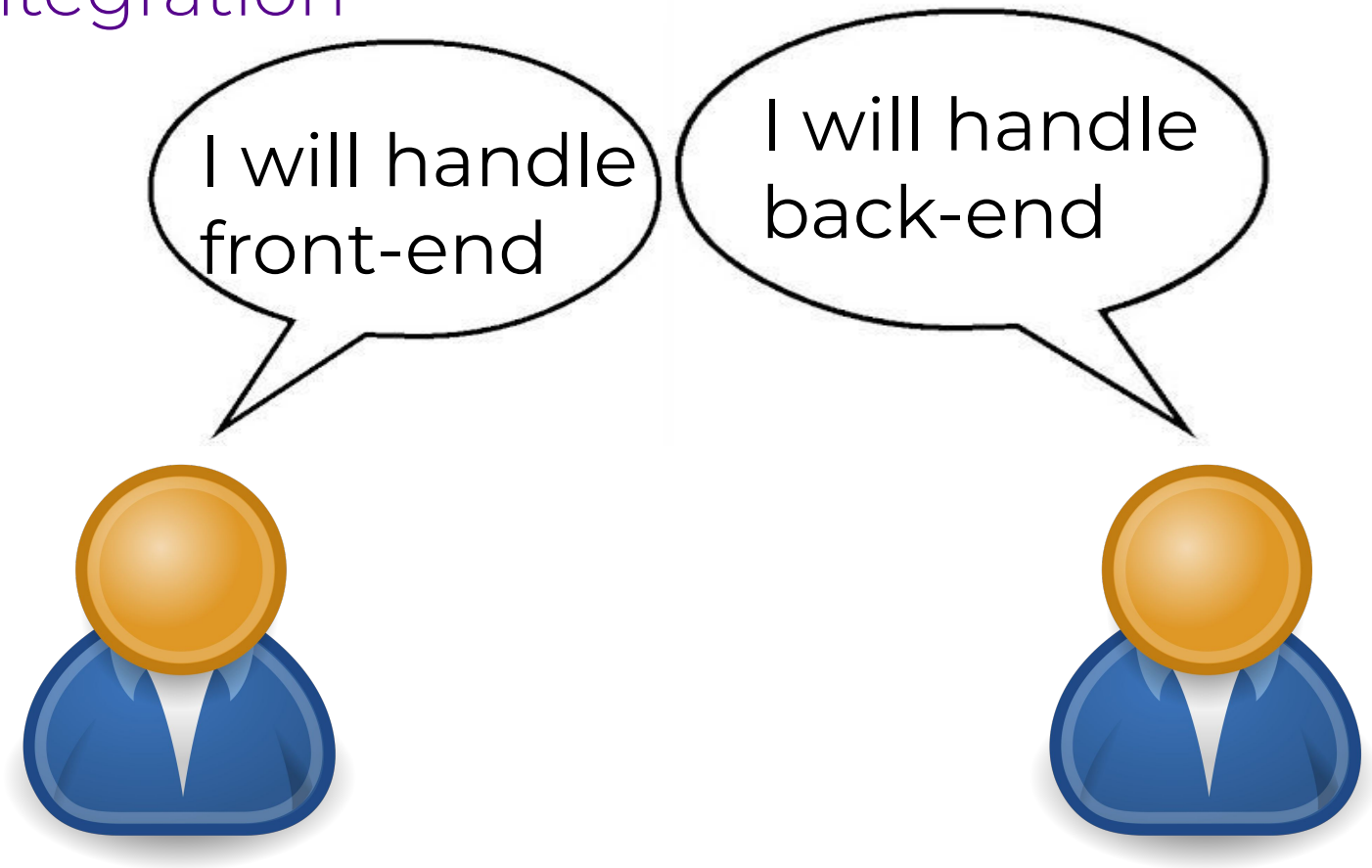
How do we avoid that?

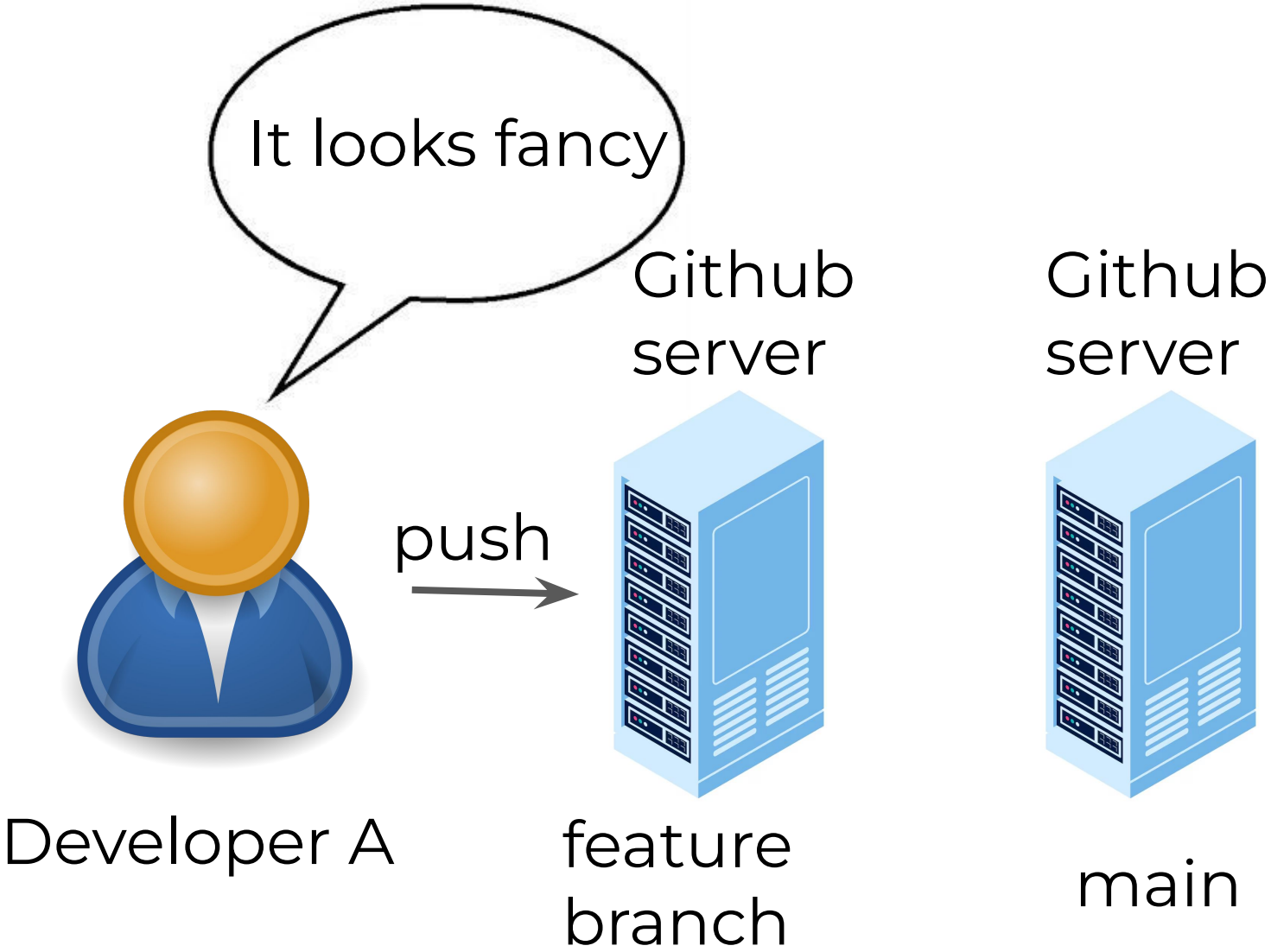
How do we avoid that?

Continuous integration

What is Continuous integration?

Let's see an example with continuous integration

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feature branch
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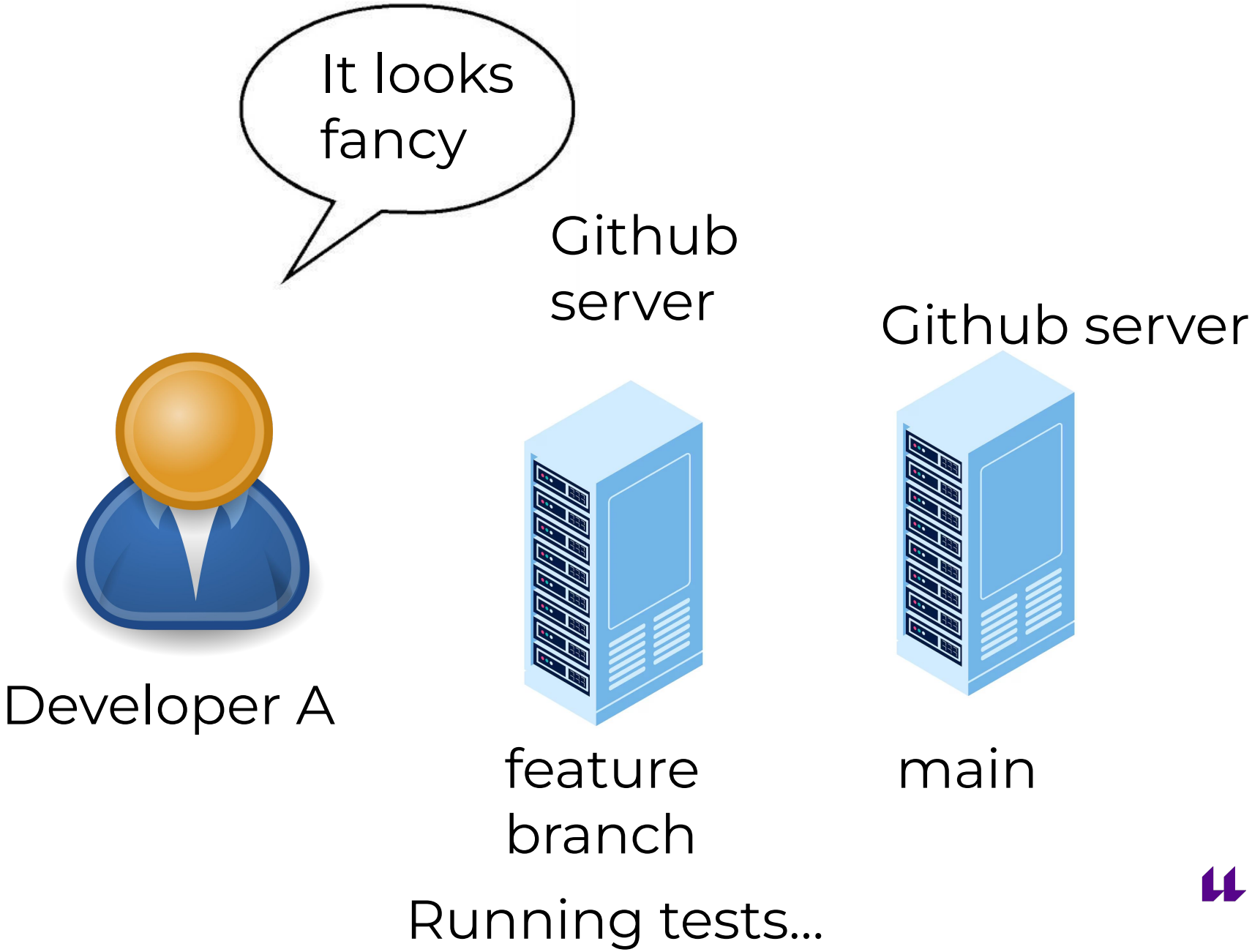
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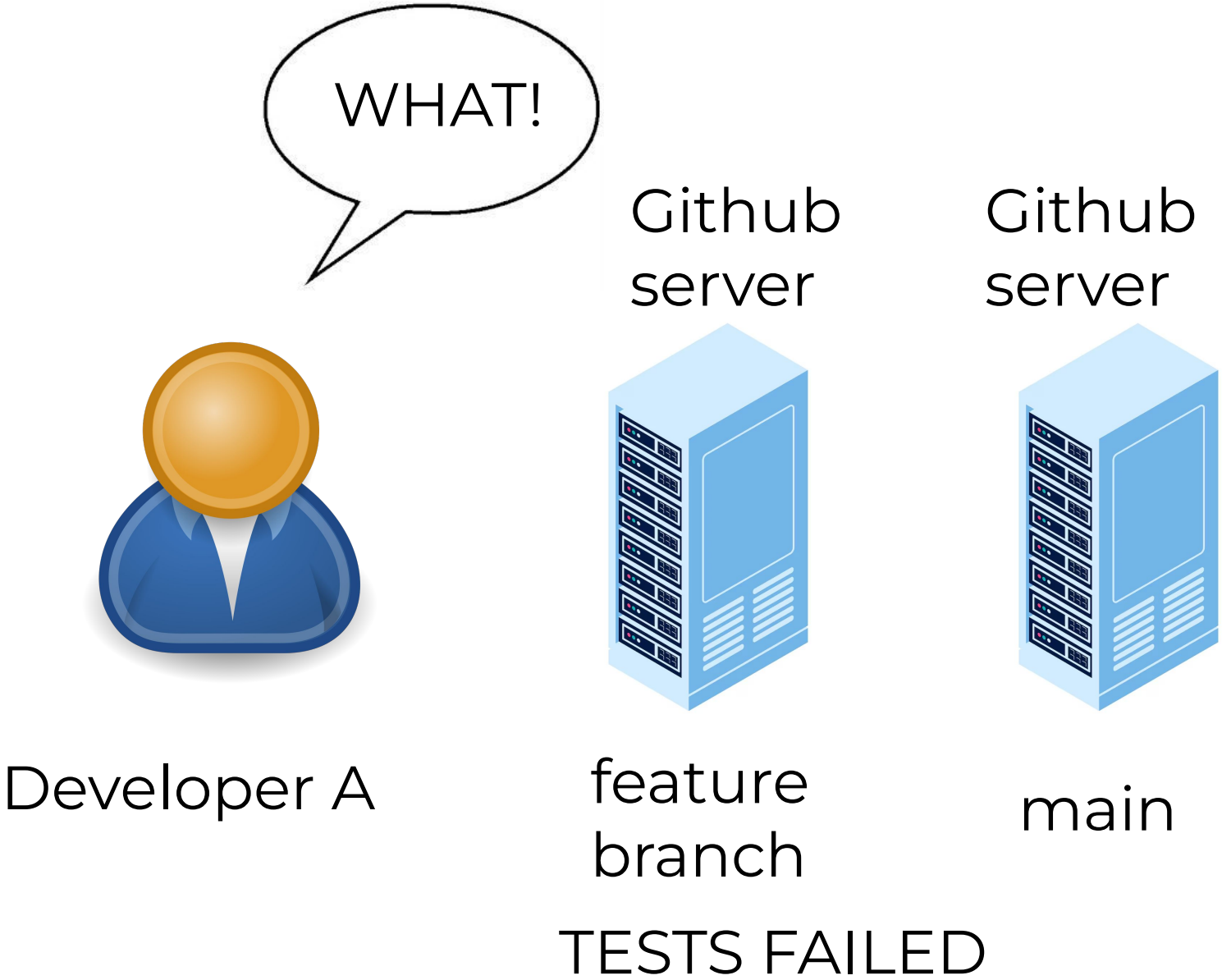
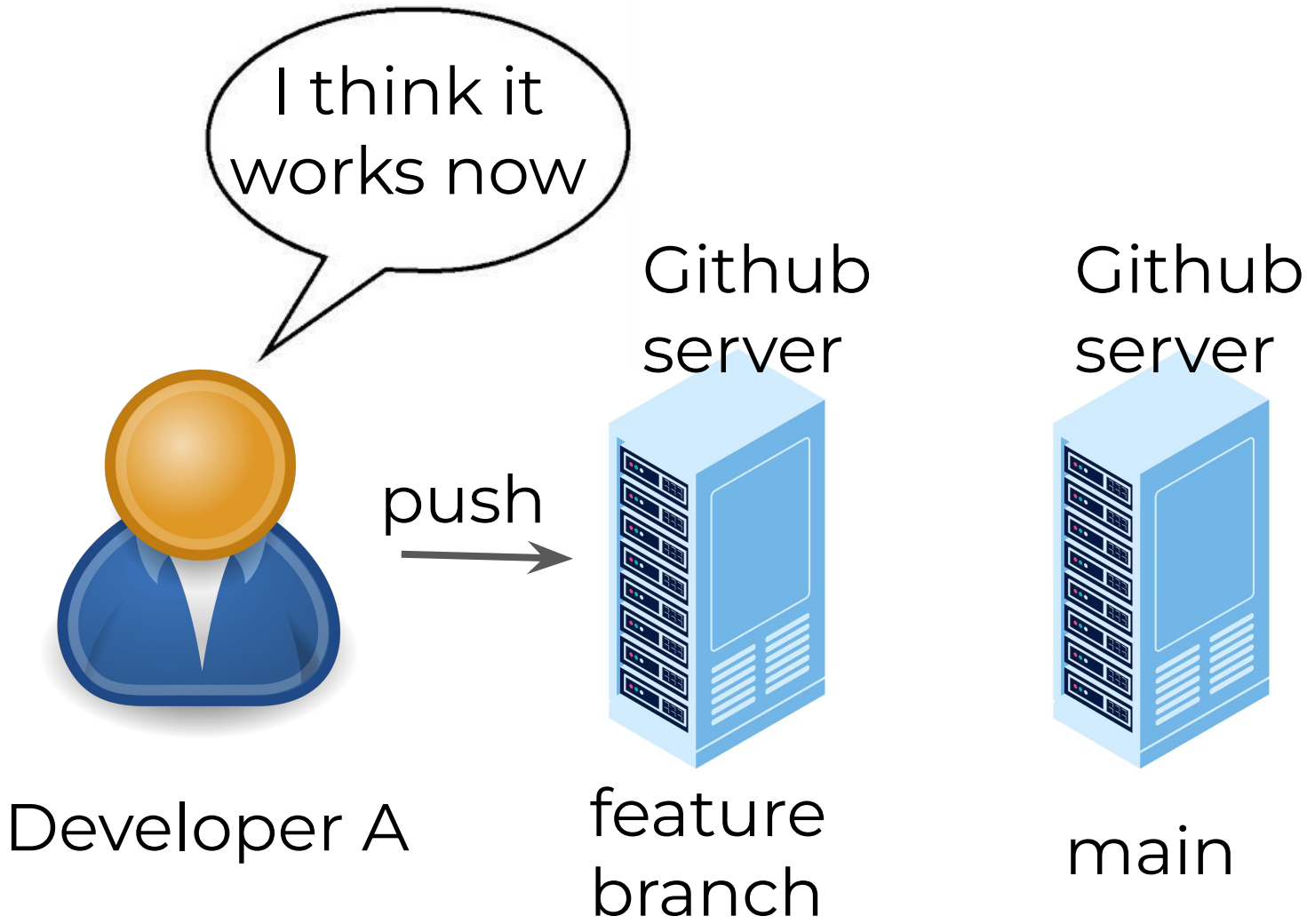
Running tests...

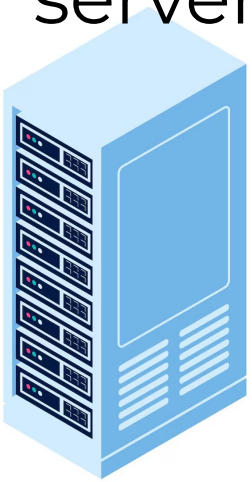

WHAT!
Github server
Github server
Developer A
feature branch
main
TESTS FAILED



Running tests...

Developer A

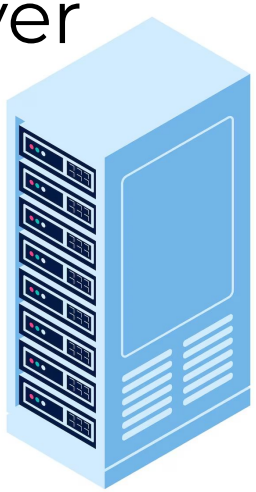
Github server



feature branch

merge

Github server



main

Tests succesful

Developer A

Developer B

Vite

- Modern frontend build tool
- Focused on **speed and simplicity**
- Great fit for TypeScript projects

Why use Vite?

Problems with Traditional Tools:

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- Slow startup time

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- Full rebundling on every change

Why use Vite?

Problems with Traditional Tools:

- Slow startup time
- Full rebundling on every change
- Poor scalability in large projects

Key Features of Vite

- Instant server start

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- Fast Hot Module Replacement (HMR)

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Key Features of Vite

- Instant server start
- Fast Hot Module Replacement (HMR)
- Native ES Modules support
- Optimized production build
- Simple and minimal configuration

How to install

```
$ npm create vite@latest
$ npm install
$ npm run dev
```

Optional configuration file: `vite.config.ts`

The `index.html` file must be in the root of the project.

Github Actions

- Automation
- Integrated
- Event-driven

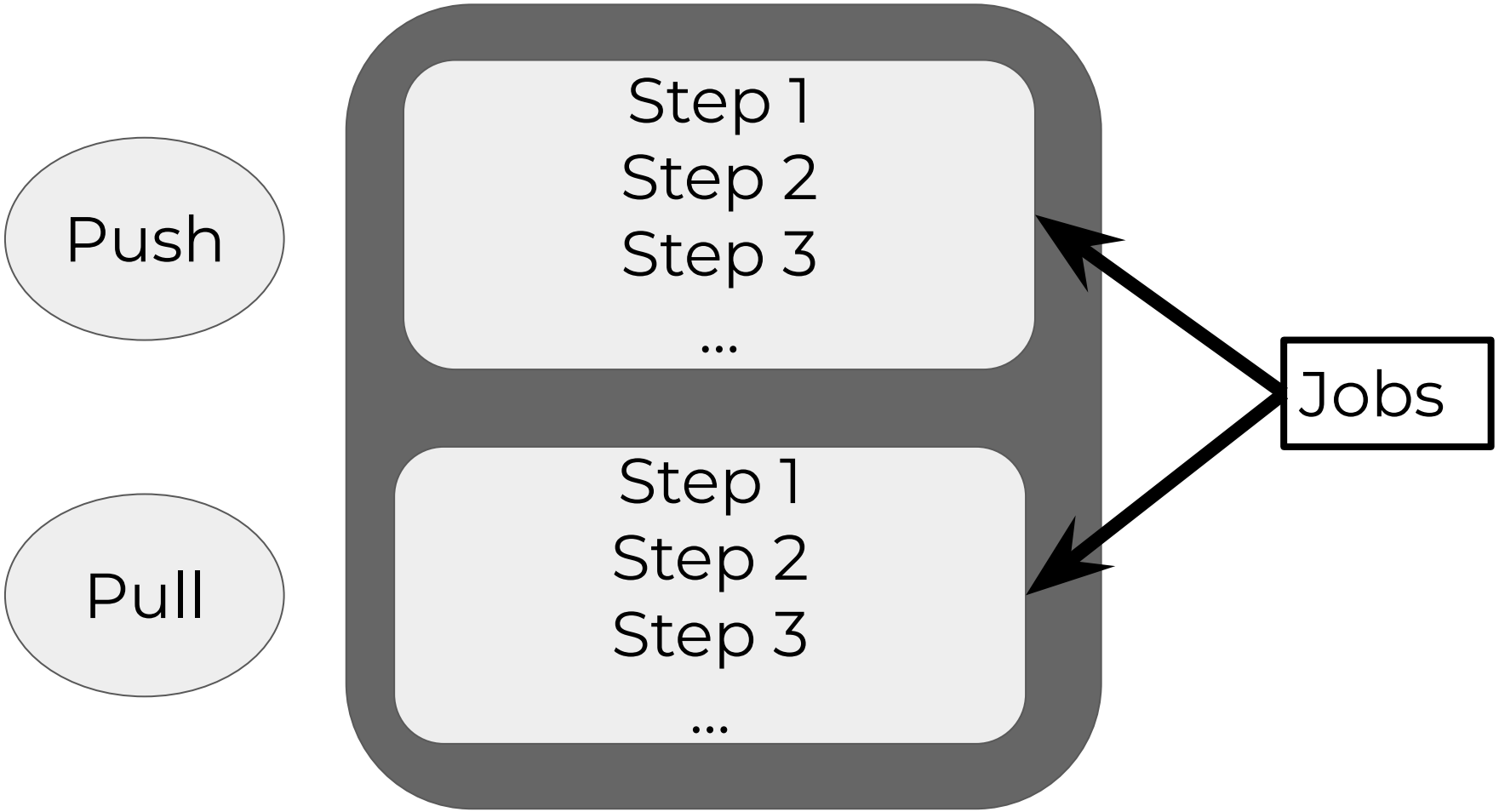
Github actions

- To use GitHub Actions, you need one or more YAML files.

EVERY `.yml` is executed automatically inside `.github/workflows`

- triggered by certain events(pull, push, etc...)

Github actions



Github actions

What do we want to add inside `.github/workflows`?

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Github actions

What do we want to add inside `.github/workflows`?

We are going to create a workflow that:

- Automatically runs our code
- Installs project dependencies
- Executes tests
- Detects errors before merging

General structure of a workflow

Main elements:

- name
- on
- jobs
- uses
- runs-on
- steps

```
name: CI Pipeline
on: [push, pull_request]
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout code
        uses: actions/checkout@v4
      - run: npm install
       - run: npm test
```

“main” and “on” elements

`name`: Name of the workflow

`on`: When the workflow runs:

- `push`
- `pull_request`
- `workflow_dispatch`

“jobs” and “uses” elements

`jobs`: Defines the set of independent or interconnected tasks.

`uses`: Runs a pre-made action created by GitHub, the community, or your team.

“runs-on” and “steps” elements

`runs-on`: Where the job runs (Ubuntu, Windows, or macOS)

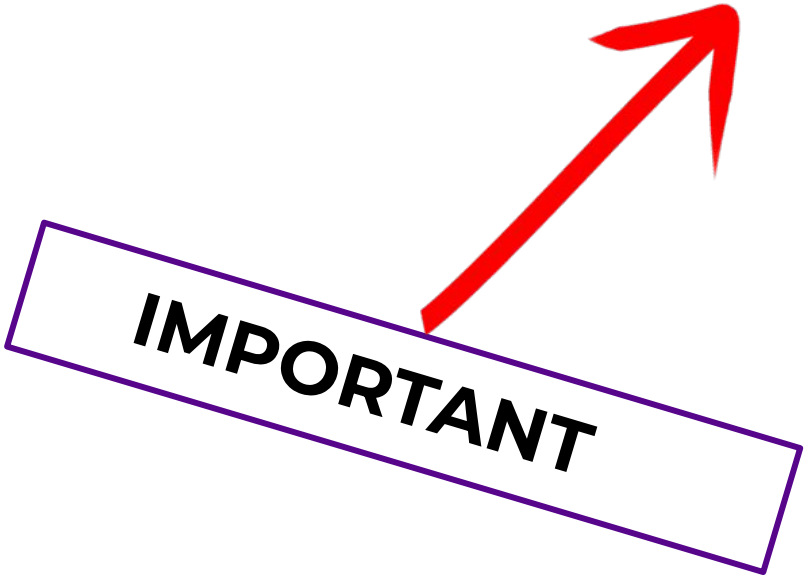
`steps`: Defines the sequence of individual commands or actions that execute in order within a job.

Github Actions Examples:
github-actions-fundamentals

Workflow

Let's start by making a feature branch

```
$ git checkout -b feature/ci-cd-pipeline
```



Continuous Delivery

- Code is always in a deployable state

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- Build, test, deploy to staging (all automated)

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- Code is always in a deployable state
- Build, test, deploy to staging (all automated)
- Production deployment requires a human click
- QA team validates before going live

Continuous Deployment

- Ship features to users as fast as possible

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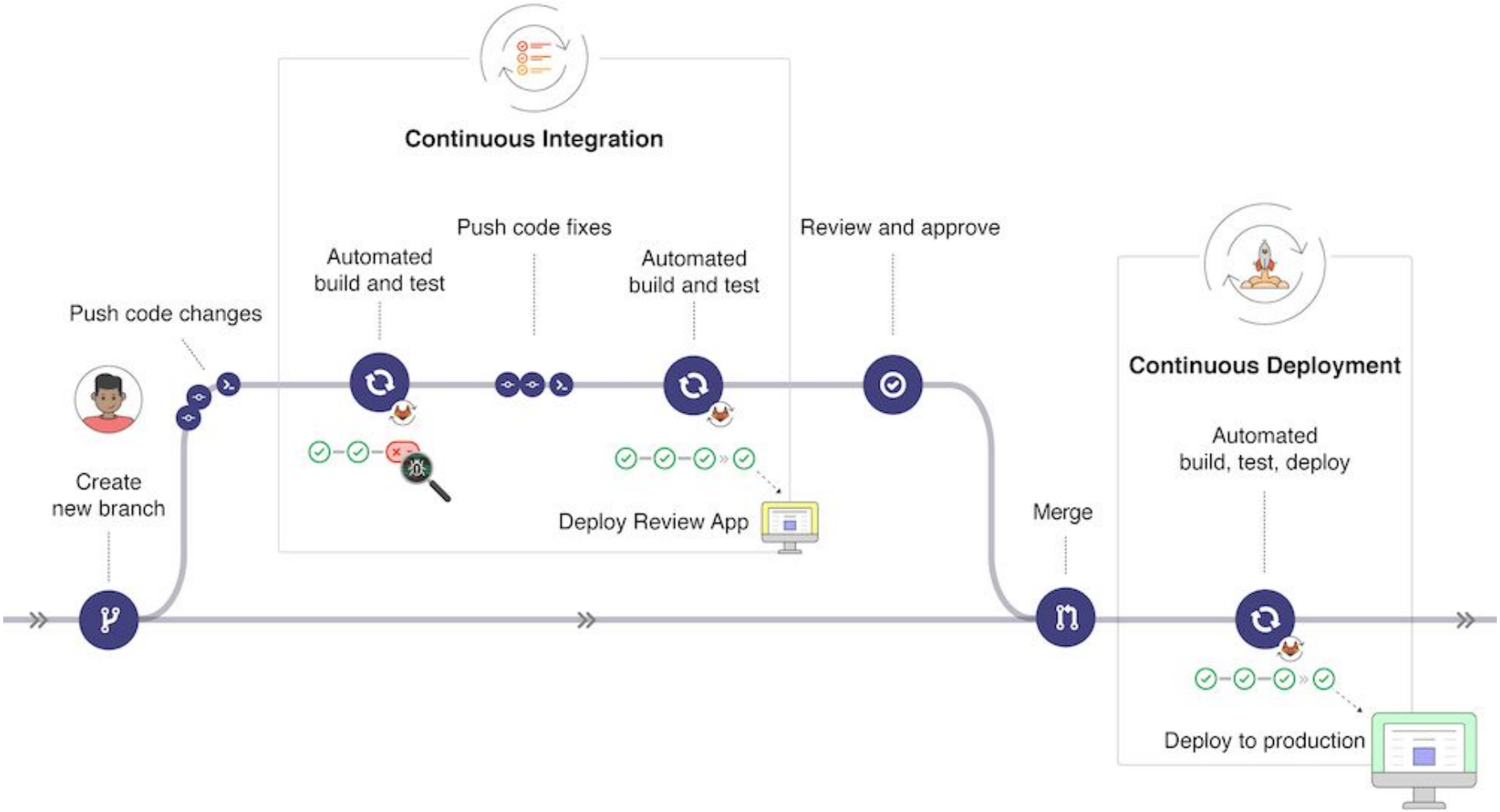
- Users get updates immediately

Continuous Deployment



Github pages

- Some key characteristics:
  - Static website hosting service

Github pages

- Some key characteristics:
  - Static website hosting service
  - Directly from a GitHub repository

Github pages

- Some key characteristics:
  - Static website hosting service
  - Directly from a GitHub repository
  - Free and easy to use

Github pages

- How does it work?
  - Push code to repository
  - GitHub selects a branch (usually main or gh-pages)
  - Static files are served as a website

Github pages

- How does it work?
  - Push code to repository
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- Deployment process with Vite
  - Build project locally (`npm run build`)
  - Generate static files (dist/)
  - Upload files to repository branch (gh-pages)

Let's see an example: Repository

Github pages

- Github Pages are usually combined with Github Actions, allowing automatic build and/or deployment.

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  - GitHub Pages = hosting service
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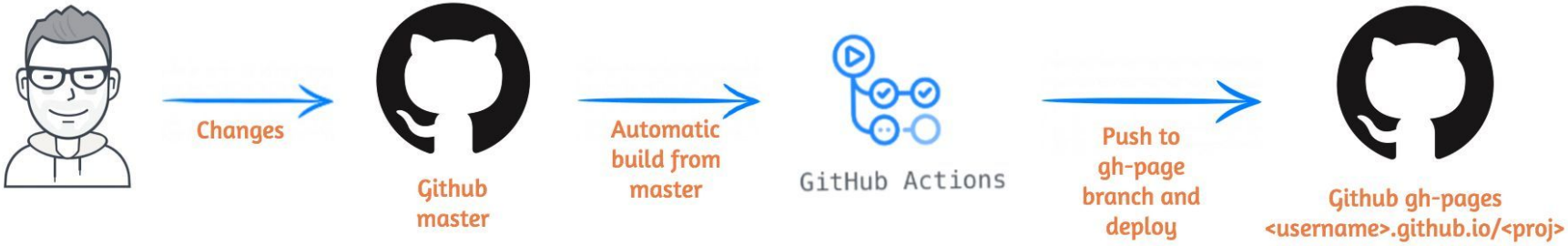

Monorepos

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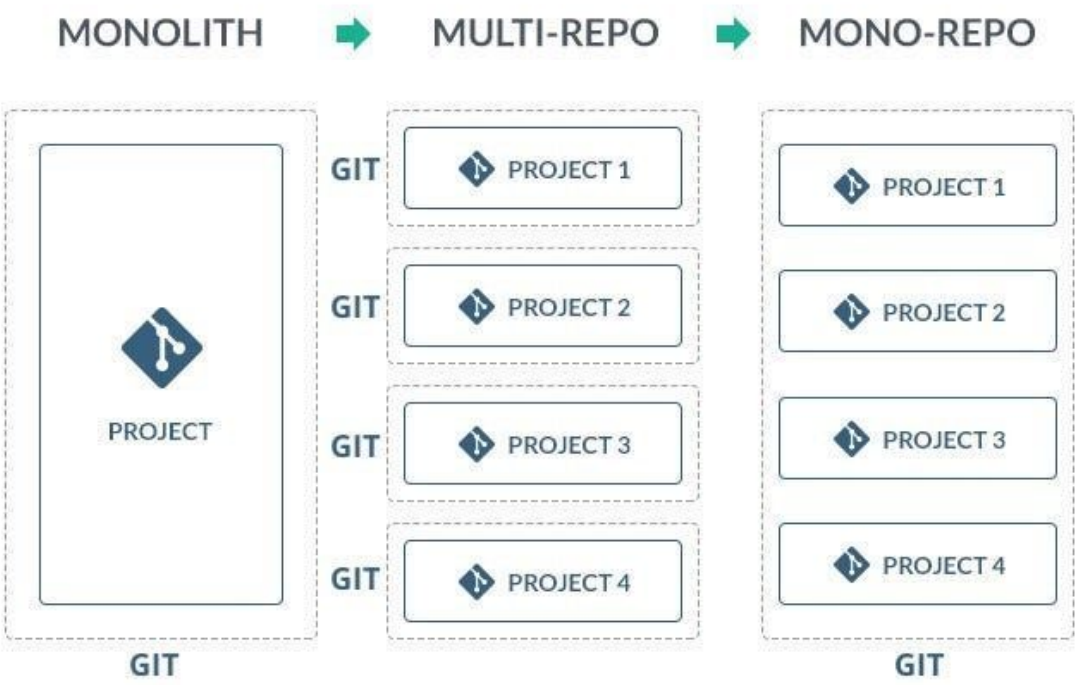

Monorepos

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- Add the repositories with:

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$ git remote add <name> <HTTP link>
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main —-squash
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Monorepos

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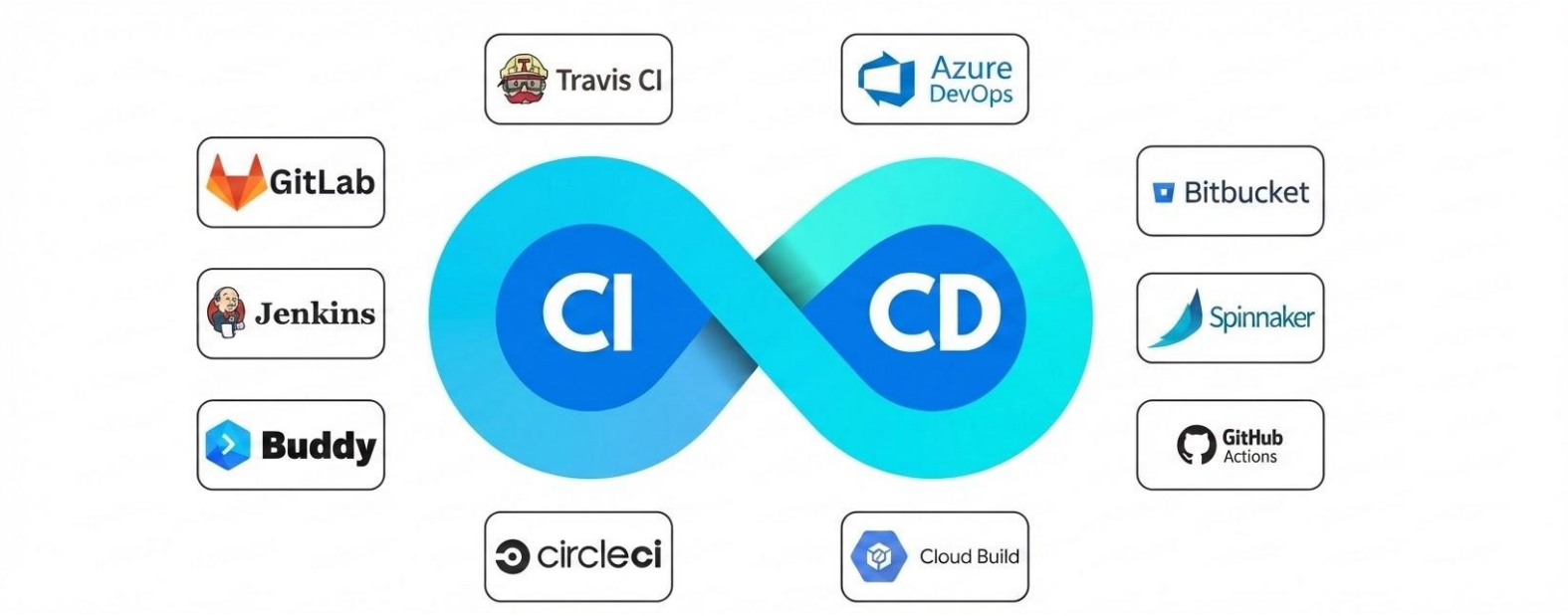
main —-squash
```

- Configure `package.json` and `vite.config.ts`

Let’s see an example: PAI-Monorepo

CI/CD Alternatives

- Some popular alternatives to GitHub Actions:
  - Jenkins
  - GitLab CI/CD
  - CircleCI
  - Google Cloud Run



CI/CD Alternatives - Jenkins

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  - Highly customizable (plugins)
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  - Requires maintenance

Jenkins

CI/CD Alternatives - GitLab CI/CD

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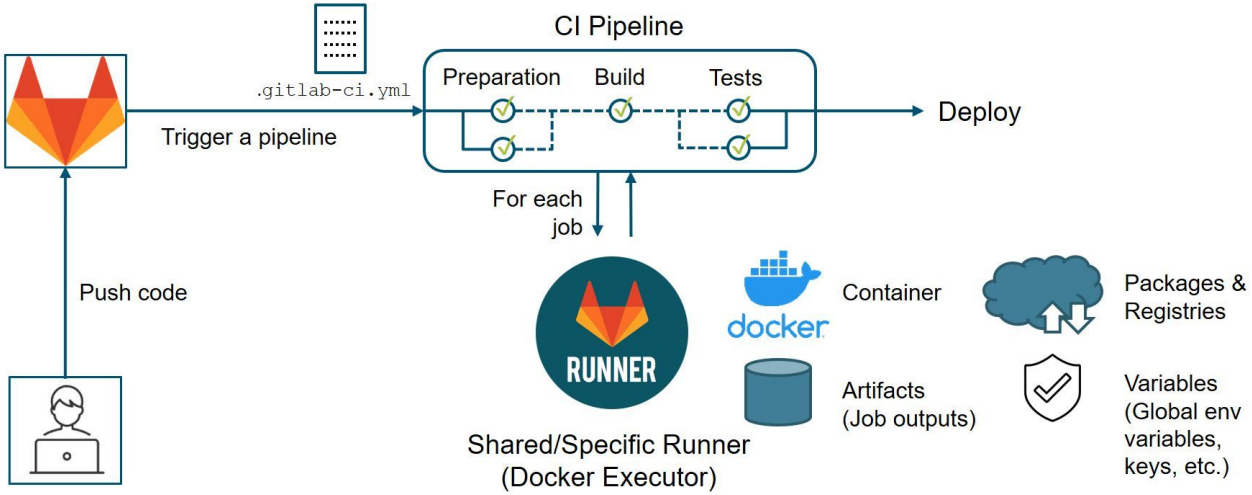
- Advantages
  - All-in-one platform
  - Easy integration



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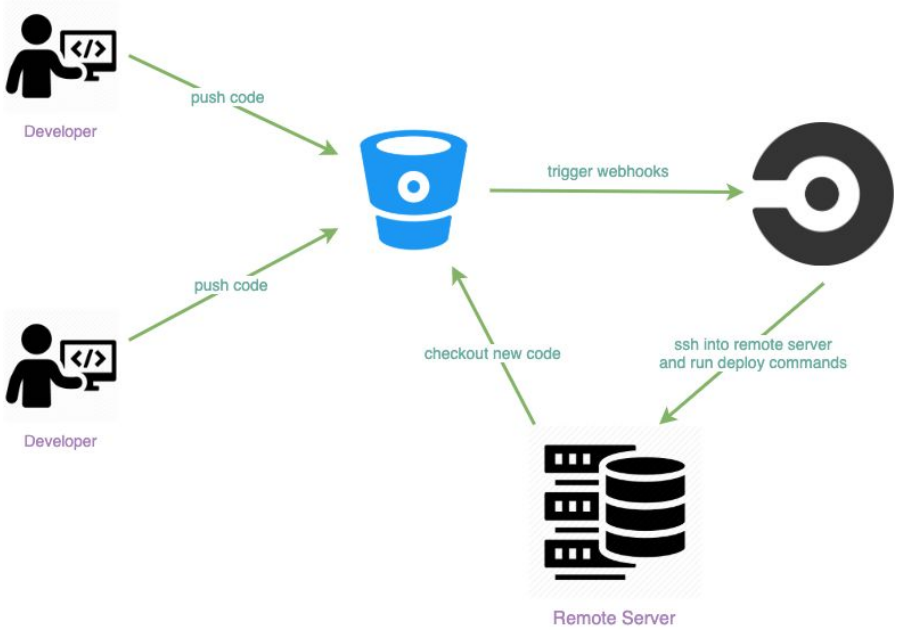
- Advantages
  - Easy to use
  - High performance



CI/CD Alternatives - Google Cloud

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- Some key characteristics:
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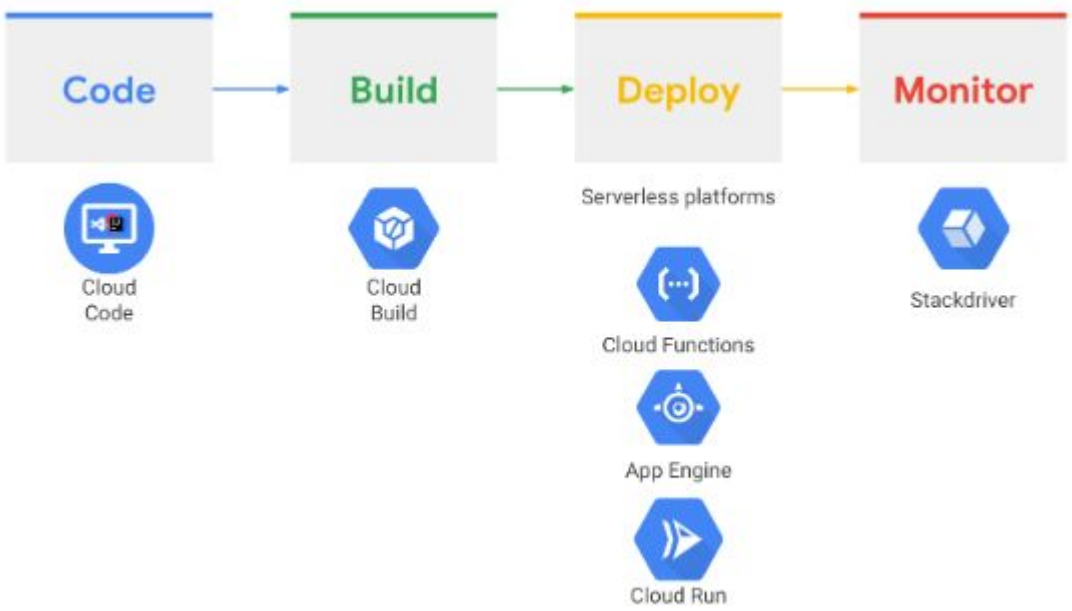
CI/CD Alternatives - Google Cloud

- Some key characteristics:
  - Paid cloud platform
  - Runs containerized applications
  - Used for deployment, not only CI

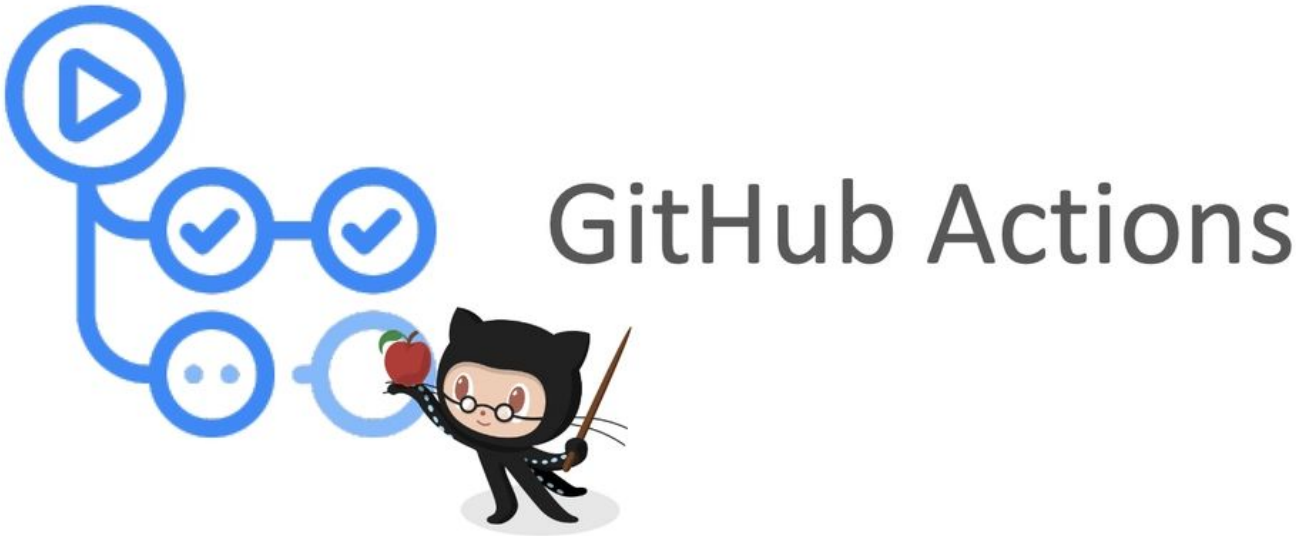


CI/CD Alternatives - Comparison

Tool	Type	Setup	Flexibility
Github Actions	Cloud	Easy	Medium
Jenkins	Self-hosted	Hard	High
GitLab CI/CD	Integrated	Medium	High
CircleCI	Cloud	Easy	Medium
Google Cloud	Cloud (Paid)	Medium	High

So, why GitHub Actions?

- Native GitHub integration
- No installation required
- Free tier available
- Simple workflows with YAML



References

- Learn Continuous Integration and Delivery with GitHub Actions, Docker, and Google Cloud Run
- GitHub Actions
- GitHub Actions TUTORIAL
- Tutorial Vite.js
- Setting up a TypeScript project with Vite
- Step-by-Step Guide: Deploy Your Vite Project to GitHub Pages with gh-pages

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